Boolean algebra and switching functions solutions

1

The terms $\bar{A}BC, A\bar{B}C, ABC$ accept exactly $011, 101, 111$, giving indices $3, 5, 7$. Combining $3, 7$ gives BC; combining $5, 7$ gives AC. Thus $f = AC + BC$.

These are all the primes. Rows $3, 5$ differ in two bits, so they do not form a two-cell cube. Every single literal includes an OFF row. Only prime BC covers m_3, and only AC covers m_5, making both essential. Any SOP can expand its terms to primes without raising cost. At least two terms and four literals are therefore required, and the displayed SOP attains both bounds.

By distributivity, $(A + B)C$ is equivalent and has three literal occurrences. It is not SOP, so it cannot establish an SOP minimum of three literals.

2

The canonical DNF is $\bar{A}B + A\bar{B}$, and the canonical CNF is $(A + B)(\bar{A} + \bar{B})$.

AB	$\bar{A}B$	$A\bar{B}$	**DNF**	$A + B$	$\bar{A} + \bar{B}$	**CNF**
00	0	0	0	0	1	0
01	1	0	1	1	1	1
10	0	1	1	1	1	1
11	0	0	0	1	0	0

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XOR is symmetric, so reversing the order leaves the true indices $1, 2$ unchanged. A general function need not be symmetric. For $f(A, B) = A\bar{B}$, the true index is 2 in order AB but 1 in order BA. Indices depend on variable order.

3

$$
\begin{aligned}
E &= (A + B)(A + \bar{B})(\bar{A} + B) \\
&= (A + B\bar{B})(\bar{A} + B) && \text{distributivity} \\
&= A(\bar{A} + B) && \text{complement and identity} \\
&= A\bar{A} + AB && \text{distributivity} \\
&= AB. && \text{complement and identity}
\end{aligned}
$$

Interchanging AND and OR in the original expression gives

$$E^d = AB + A\bar{B} + \bar{A}B = A + B.$$

The last equality follows by dualizing the entire derivation, or directly: $AB + A\bar{B} = A$, and $A + \bar{A}B = (A + \bar{A})(A + B) = A + B$.

Boolean algebra and switching functions solutions (continued)

Applying De Morgan to the original expression gives

$$\bar{E} = \bar{A}\bar{B} + \bar{A}B + A\bar{B} = \bar{A} + \bar{B} = \overline{AB}.$$

Complementing negates each literal as well as interchanging the operations. Its result differs from the dual $A + B$.

4

Choose output 1 on both don't-care rows $1, 5$. The group $\{0, 1, 4, 5\}$ gives $\bar{B}$; the group $\{0, 2\}$ gives $\bar{A}\bar{C}$. Thus

$$g = \bar{B} + \bar{A}\bar{C}.$$

Index	0	1	2	3	4	5	6	7
Specification	ON	D	ON	OFF	ON	D	OFF	OFF
g	1	1	1	0	1	1	0	0

The only legal one-literal term is $\bar{B}$, which misses 2; every other single literal hits an OFF row. With only two literal occurrences, two one-literal terms merely repeat $\bar{B}$. A single two-literal term cannot accept all of $000, 010, 100$, whose only common fixed bit is $C = 0$. Neither constant works. Thus at least three literals are needed, and the expression attains the bound.

5

The four minterms are $\bar{A}\bar{B}\bar{C}, \bar{A}\bar{B}C, \bar{A}B\bar{C}, \bar{A}BC$. All occupy row $A = 0$, with columns ordered $BC = 00, 01, 11, 10$. Algebraically,

$$\begin{aligned} f &= \bar{A}(\bar{B}\bar{C} + \bar{B}C + B\bar{C} + BC) \\ &= \bar{A}\{\bar{B}(\bar{C} + C) + B(\bar{C} + C)\} \\ &= \bar{A}(\bar{B} + B) = \bar{A}. \end{aligned}$$

The steps use distributivity, complements and identities.

With rows $AB = 00, 01, 11, 10$, the cells $\{0, 1\}$ occupy the top row and $\{4, 5\}$ the bottom. The row labels $00, 10$ differ only in A, giving wrap-around adjacency. In the combined group, A, C are free and only $B = 0$ stays fixed, so $\Sigma m(0, 1, 4, 5) = \bar{B}$. The two diagrams use different axes; the first map's row labels do not apply to the second.

Boolean algebra and switching functions solutions (continued)

6

An atom is a minimal nonzero element. If $0 = 1$, every x is the unique element 1. There are no atoms, and both the premise and conclusion hold.

Suppose $0 \neq 1$. Every nonzero y in a finite algebra has an atom below it: choose a minimal element of the finite nonempty set $\{z : 0 < z \leq y\}$. Let s be the finite join of all atoms. If $s \neq 1$, then $\bar{s} \neq 0$, so it has an atom a below it. But $a \leq s$ because it is one of the atoms, and $a \leq \bar{s}$, forcing $a = 0$, a contradiction. Hence $s = 1$.

Every atom lies below x by hypothesis. The least-upper-bound property yields $1 = s \leq x$. Since $x \leq 1$, antisymmetry gives $x = 1$. A nonzero finite algebra cannot have no atoms, because taking $y = 1$ would produce one. This covers every possible atomless case.

7

All primes are listed below.

Symbol	Prime	Covered indices
p_1	$\bar{A}C$	1,3
p_2	$\bar{A}B$	2,3
p_3	$B\bar{C}$	2,6
p_4	$A\bar{C}$	4,6
p_5	$A\bar{B}$	4,5
p_6	$\bar{B}C$	1,5

Every single literal includes 000 or 111, and is illegal. Each ON row has exactly two ON neighbors differing in one bit, so the table lists every two-cell cube. Three-literal singletons expand and are not prime. Each ON row belongs to two primes, so there are no essential primes.

Six ON rows must be covered, and each term covers at most two. At least three terms are required. A three-term cover can have no overlap, or its union would contain fewer than six rows. On the cycle $1 - 3 - 2 - 6 - 4 - 5 - 1$, three disjoint edges must alternate. Thus all prime covers of minimum term count are

$$\{p_1, p_3, p_5\}, \qquad \{p_2, p_4, p_6\}.$$

Boolean algebra and switching functions solutions (continued)

They give $\bar{A}C + B\bar{C} + A\bar{B}$ and $\bar{A}B + A\bar{C} + \bar{B}C$. Each legal term needs at least two literals, so any three-term solution needs at least six. Both achieve six. More terms need at least eight literals, so these also minimize total literal count globally.

8

Set

$$n_B = \mathrm{NAND}(B, B), \quad t_1 = \mathrm{NAND}(A, n_B), \quad t_2 = \mathrm{NAND}(B, C), \quad f = \mathrm{NAND}(t_1, t_2).$$

De Morgan gives $f = A\bar{B} + BC$.

ABC	n_B	t_1	t_2	f
000	1	1	1	0
001	1	1	1	0
010	0	1	1	0
011	0	1	0	1
100	1	0	1	1
101	1	0	1	1
110	0	1	1	0
111	0	1	0	1

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The longest path $B \to n_B \to t_1 \to f$ has three levels. The other branch $B, C \to t_2 \to f$ has two. If $\bar{B}$ is already provided, omit the inverter NAND; the remaining three gates have maximum depth two. This is a feasible construction and does not prove a globally minimal gate count.

9

The representation theorem gives $B \cong \mathcal{P}(A)$, with $2^{|A|} = 16$, so there are four atoms.

Fix an atom a. The condition $a \leq x$ becomes membership of a in $\Phi(x)$. The other three atoms may each be included or excluded, giving $2^3 = 8$ elements above it.

The condition $x \wedge y = 0$ becomes $\Phi(x) \cap \Phi(y) = \varnothing$. Each atom has three allowed locations: in x only, in y only, or in neither. It cannot be in both. Independent choices for four atoms give $3^4 = 81$ ordered pairs. Pairs with $x = 0$ or $y = 0$ are included.

Boolean algebra and switching functions solutions (continued)

10

The prime-factor-subset correspondence gives

$$\bar{6} = 5, \qquad 6 \wedge 10 = \gcd(6, 10) = 2, \qquad 6 \vee 10 = \operatorname{lcm}(6, 10) = 30.$$

An automorphism preserves order and nonzero elements, hence permutes atoms $2, 3, 5$. Conversely, any permutation of the atoms extends to an automorphism of subsets and their operations. Each element is its atom join, so the extension is unique.

To send $6 \leftrightarrow \{2, 3\}$ to $10 \leftrightarrow \{2, 5\}$, the remaining atom 5 must map to 3, and $2, 3$ must map bijectively to $2, 5$. Exactly two permutations work:

- $2 \mapsto 2,\ 3 \mapsto 5,\ 5 \mapsto 3.$
- $2 \mapsto 5,\ 3 \mapsto 2,\ 5 \mapsto 3.$

The images of other divisors are products of the corresponding distinct prime images. For example, the second map sends $10 = 2 \cdot 5$ to $5 \cdot 3 = 15$.

For divisors of 12, the proposed complement $12/2 = 6$ fails both tests: $\gcd(2, 6) = 2 \neq 1$ and $\operatorname{lcm}(2, 6) = 6 \neq 12$. In fact 2 has no complement: only divisors $1, 3$ are coprime to it, and neither has lcm 12 with it. The construction for 30 depends on its being square-free.

11

Three atoms a, b, c force the eight distinct joins

$$0,\ a,\ b,\ c,\ a \vee b,\ a \vee c,\ b \vee c,\ a \vee b \vee c.$$

If two different subsets of atoms had the same join, choose an atom appearing on only one side and meet both sides with it. The side containing it gives that atom. The other gives 0, because distinct atoms have meet zero. This contradicts its being nonzero. Thus three atoms already force at least eight elements, ruling out six. For a finite algebra with exactly three atoms, representation gives exactly eight.

For the infinite counterexample, use the circle $[0, 1)$, with $0, 1$ identified. Take all finite unions of half-open arcs whose endpoints are dyadic fractions $k/2^r$, including the empty set. Finitely many endpoints lie in one common grid with spacing 2^{-r}. Each element is a union of grid cells, and intersections, unions and complements remain such unions, verifying a Boolean algebra.

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Boolean algebra and switching functions solutions (continued)

Every nonempty element contains a cell. Bisect that cell and take one half. The endpoints remain dyadic, and the result is a nonempty element strictly smaller than the original. There are no atoms. Yet the empty set differs from the full circle, so the algebra is nonzero. The finite theorem does not apply.

12

For every x,

$$(x + x)^2 = x + x.$$

Distributivity and $x^2 = x$ make the left side $4x$, so $4x = 2x$. Additive cancellation gives $2x = 0$. In particular, $2 \cdot 1 = 0$. If $1 \neq 0$, the characteristic is 2; if $1 = 0$, this is the zero ring, whose characteristic is 1 under Chapter 14's convention.

Expanding $(x + y)^2 = x + y$ and cancelling $x^2 = x, y^2 = y$ yields $xy + yx = 0$. Each element is its own additive inverse, so $yx = -xy = xy$, proving commutativity.

Define $x \wedge y = xy$, $\bar{x} = 1 + x$ and $x \vee y = x + y + xy$. The order and lattice verification in the lecture proves these form a Boolean algebra. Boolean XOR is

$$(x \wedge \bar{y}) \vee (\bar{x} \wedge y).$$

The two terms have product zero, so their join equals their ring sum:

$$x(1 + y) + (1 + x)y = x + xy + y + xy = x + y.$$

Thus $x \oplus y = x + y$, while OR is $x + y + xy$. They coincide exactly when $xy = 0$. In a nonzero algebra, $x = y = 1$ gives XOR 0 and OR 1, the smallest counterexample to identifying the operations.

Boolean algebra and switching functions solutions (continued)

Verification code appendix: verify_boolean.py

```
"""Verify the canonical DNF, complement, and four-cell K-map identities."""


def bits(assignment: int) -> tuple[int, int, int]:
    return (assignment >> 2) & 1, (assignment >> 1) & 1, assignment & 1


def left_dnf(a: int, b: int, c: int) -> int:
    return int(not a and b and c) | int(a and not b and c) | int(a and b and c)


def right_sop(a: int, b: int, c: int) -> int:
    return int(a and c) | int(b and c)


for assignment in range(8):
    assert left_dnf(*bits(assignment)) == right_sop(*bits(assignment))

for assignment in range(8):
    a, b, c = bits(assignment)
    original = (a or b) and c
    complement = ((not a) and (not b)) or (not c)
    if original:
        assert not complement
    else:
        assert complement

for assignment in range(8):
    a, _, _ = bits(assignment)
    kmap_function = assignment < 4
    assert kmap_function == (not a)

print("boolean examples: PASS")
```

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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